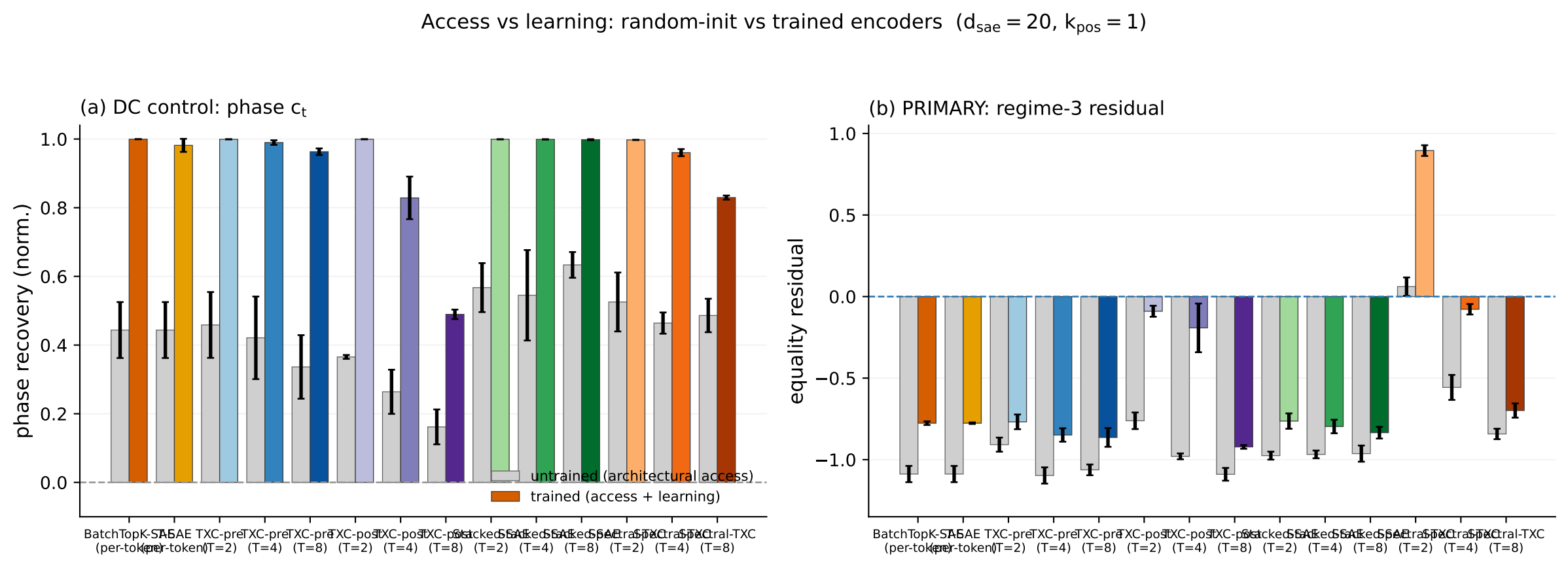

Access vs learning: random-init vs trained encoders ($d_{sae} = 20$, $k_{pos} = 1$)
(a) DC control: phase c_t
phase recovery (norm.)
1.0
0.8
0.6
0.4
0.2
0.0
untrained (architectural access)
trained (access + learning)
(b) PRIMARY: regime-3 residual
equality residual
1.0
0.5
0.0
−0.5
−1.0
BatchTopK-SAE (per-token)
TopK-SAE (per-token)
TXC-pre (T=2)
TXC-pre (T=4)
TXC-pre (T=8)
TXC-post (T=2)
TXC-post (T=4)
TXC-post (T=8)
Stacked SAE (T=2)
Stacked SAE (T=4)
Stacked SAE (T=8)
SAE (T=2)
SAE (T=4)
SAE (T=8)
SPECtral-TXC (T=2)
SPECtral-TXC (T=4)
SPECtral-TXC (T=8)